

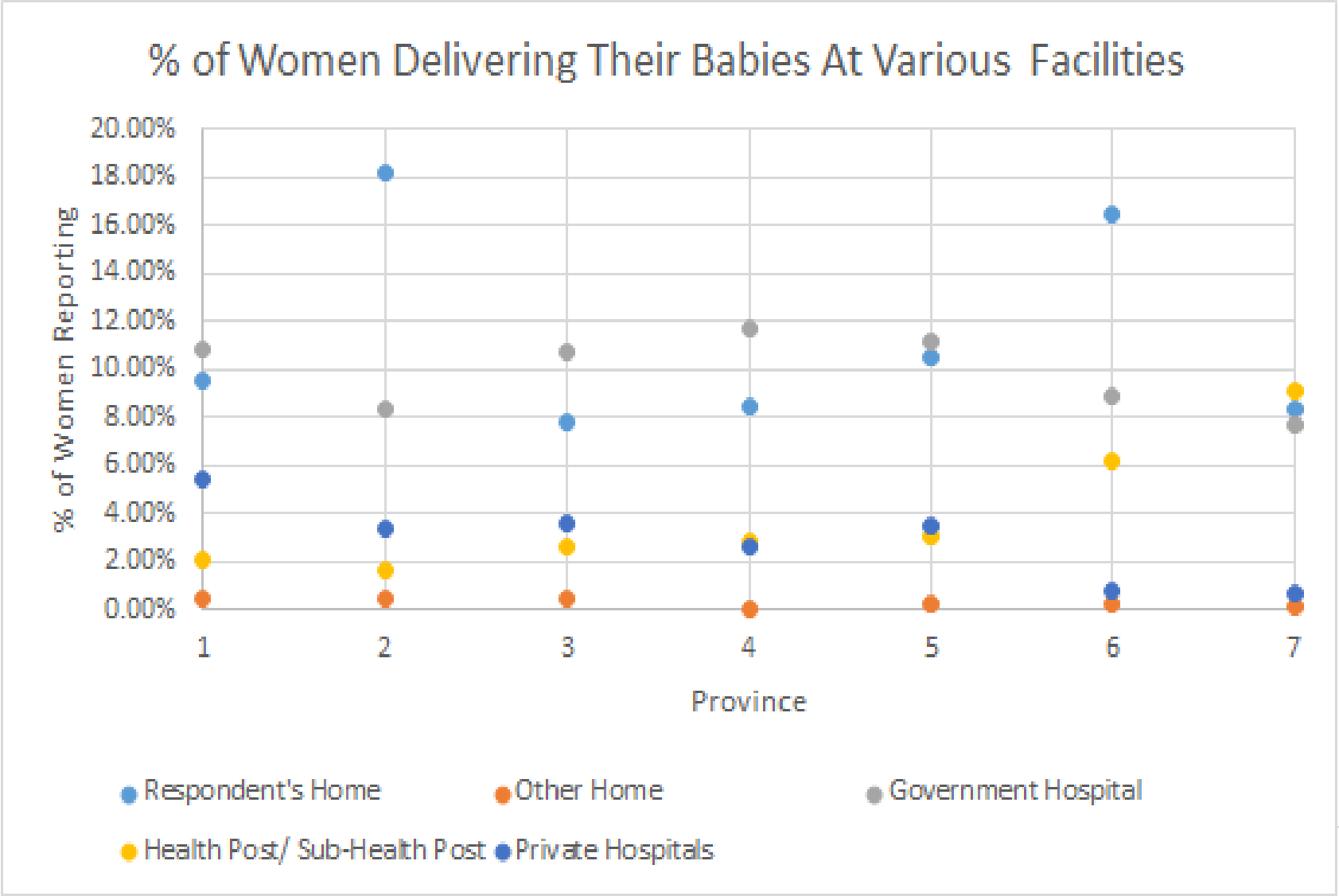
The Impact of Socioeconomic Status on Maternal Healthcare Utilization in Nepal: Evidence from the Nepal Demographic Health Survey 2016

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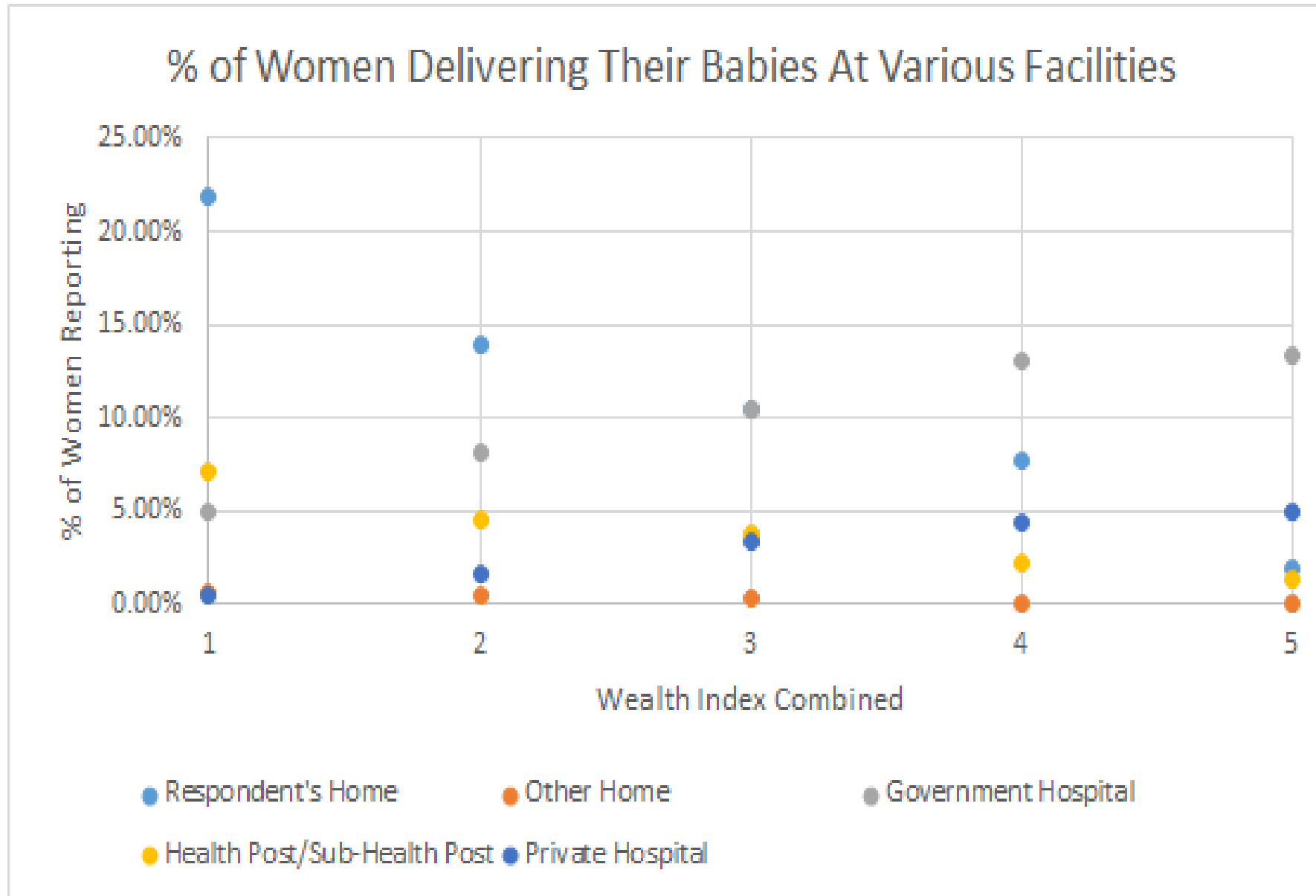
ECON 580:Honors Research Tutorial

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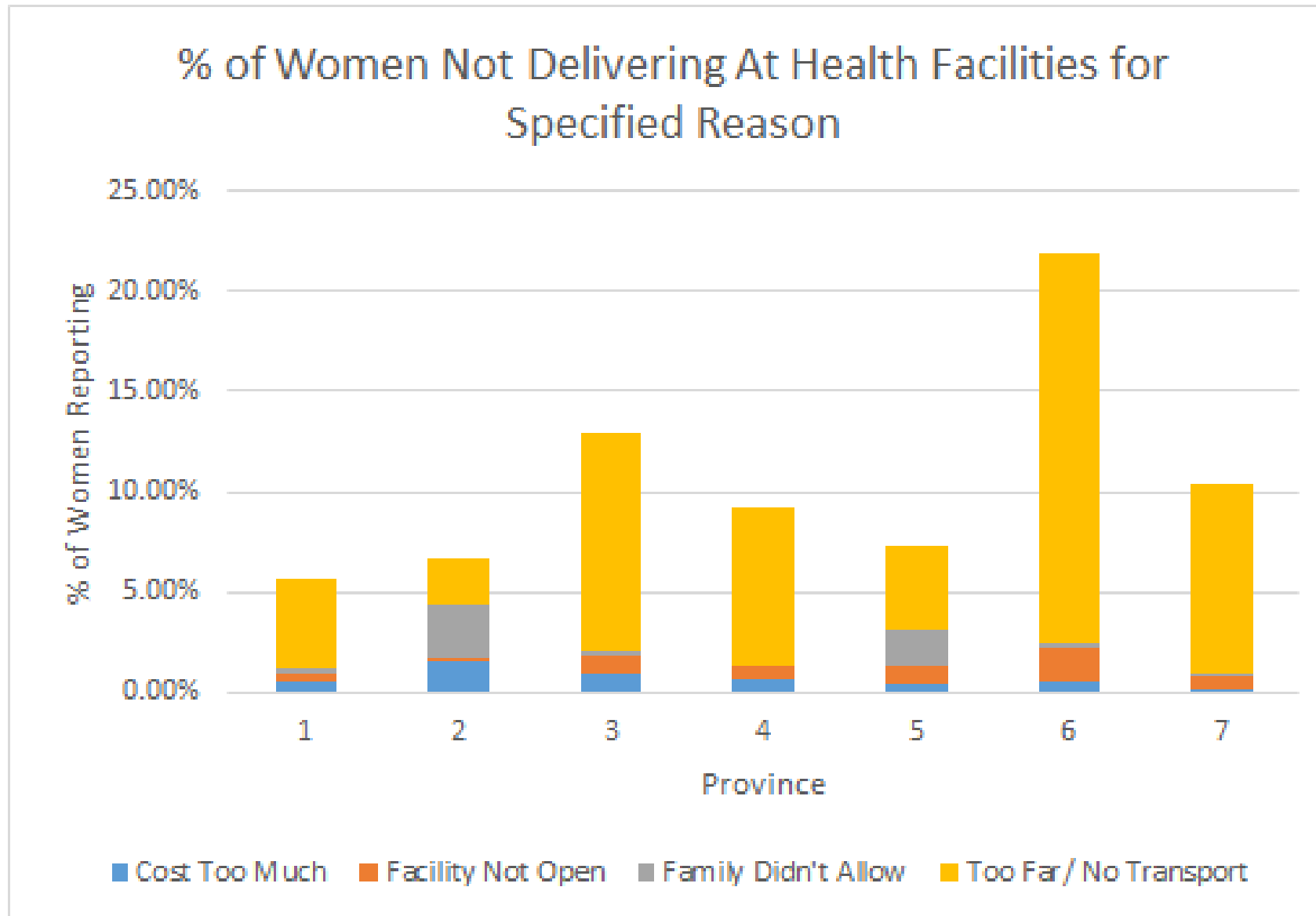
Visualize Motivation?



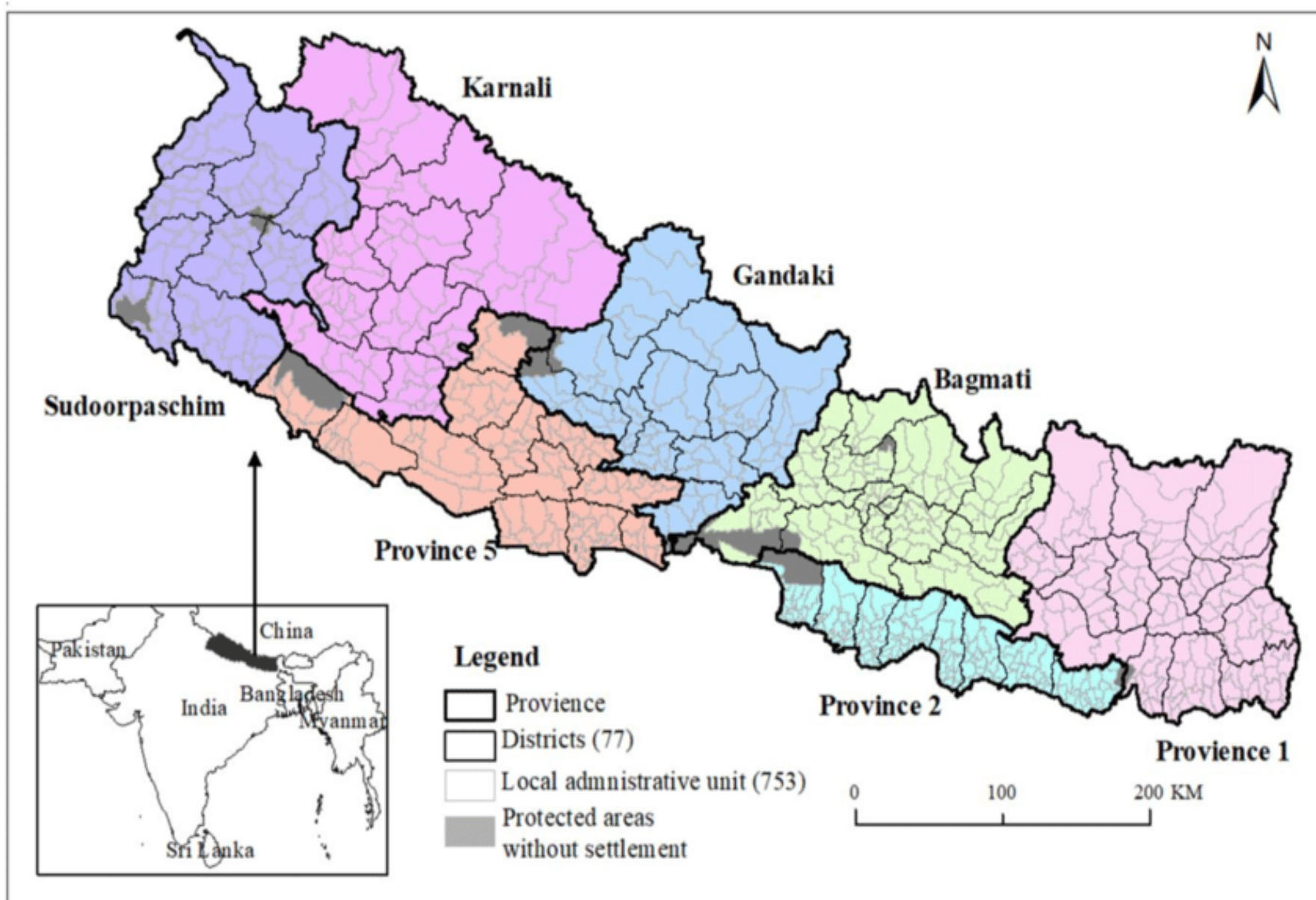
Visualize Motivation contd?



What might be the REASON?



Where are these Provinces?



Motivation, Question & Contribution

Motivation:

- ❖ Maternal healthcare utilization is a critical determinant of maternal and child health outcomes, particularly in low-income countries such as Nepal.
- ❖ Despite improvements in health infrastructure, substantial disparities persist in access to essential maternal health services, with socioeconomic status (SES)-specifically household wealth and maternal education-emerging as major barriers.
- ❖ Understanding these disparities is essential for designing effective, equitable health interventions

How does socioeconomic status-measured by household wealth and educational attainment-affect the utilization of maternal healthcare services (antenatal care, institutional delivery, and infant survival) in Nepal?

Contributions:

- ❖ This study extends the literature by jointly analyzing the effects of wealth and education on multiple maternal health outcomes using nationally representative data.
- ❖ It further contributes by quantifying the interaction between wealth and education, and by exploring geographic disparities at the provincial level, thus informing both targeted and systemic policy responses

Data

The analysis uses the 2016 Nepal Demographic and Health Survey (NDHS), a nationally representative cross-sectional dataset. The sample includes women aged 15–49 who reported at least one pregnancy in the five years preceding the survey.

❖ Dependent Variables:

- Institutional delivery (binary: 1 if birth in a health facility, 0 otherwise)
- Number of antenatal care (ANC) visits (continuous)
- Postnatal check within 2 months (binary: 1 if yes, 0 otherwise)
- Number of infant deaths (continuous)

❖ Main Explanatory Variables:

- Wealth quintile (poorest to richest, based on asset indices)
- Educational attainment (none, primary, secondary, higher)

❖ Controls:

- Urban/rural residence, maternal age, number of children, provincial dummies

Regression Equation 1

$$ANCVisits_i = \gamma_0 + \gamma_1 WealthQuintile_i + \gamma_2 Education_i + \gamma_3 Urban_i + \gamma_4 Age_i + \gamma_5 NumberofChildren_i + \varepsilon_i$$

- ***ANCVisits_i*** = Number of antenatal care visits that mother *i* received during her most recent pregnancy.
- ***WealthQuintile_i*** = Household wealth quintile rank of mother *i* (typically an ordinal scale from 1 = poorest to 5 = richest). Higher values indicate higher socioeconomic status.
- ***Education_i*** = Education level of mother *i* (e.g. years of schooling completed). A higher value denotes more education.
- ***Urban_i*** = Urban residence indicator for mother *i* (1 if living in an urban area, 0 if in a rural area). Controls for urban–rural differences in access and behavior.
- ***Age_i*** = Age of mother *i* (in years). Included as a control.
- ***NumberofChildren_i*** = Number of children mother *i* has (parity). Controls for family size or prior births, which may affect healthcare utilization.

Note: All models include interaction terms between wealth and education, and province fixed effects to account for unobserved regional heterogeneity

Regression Equation 2

$$\log \left(\frac{p_{(ID_i=1)}}{1-p_{(ID_i=1)}} \right) = \beta_0 + \beta_1 \textit{WealthQuintile}_i + \beta_2 \textit{Education}_i + \beta_3 \textit{Urban}_i + \beta_4 \textit{Age}_i + \beta_5 \textit{NumberofChildren}_i + \varepsilon_i$$

- **ID_i** = Indicator for whether mother i delivered in a health institution (1 if yes, 0 if no).
- **$\textit{WealthQuintile}_i$** = Household wealth quintile rank of mother i (typically an ordinal scale from 1 = poorest to 5 = richest). Higher values indicate higher socioeconomic status.
- **$\textit{Education}_i$** = Education level of mother i (e.g. years of schooling completed). A higher value denotes more education.
- **$\textit{Urban}_i$** = Urban residence indicator for mother i (1 if living in an urban area, 0 if in a rural area). Controls for urban–rural differences in access and behavior.
- **$\textit{Age}_i$** = Age of mother i (in years). Included as a control.
- **$\textit{NumberofChildren}_i$** = Number of children mother i has (parity). Controls for family size or prior births, which may affect healthcare utilization.

Note: All models include interaction terms between wealth and education, and province fixed effects to account for unobserved regional heterogeneity

Regression Equation 3

$$\log \left(\frac{p(PN_i=1)}{1-p(PN=1)} \right) = \beta_0 + \beta_1 \textit{WealthQuintile}_i + \beta_2 \textit{Education}_i + \beta_3 \textit{Urban}_i + \beta_4 \textit{Age}_i + \beta_5 \textit{NumberOfChildren}_i + \varepsilon_i$$

- **PN_i** = Postnatal check within 2 months (e.g., 1 = Yes, 0 = No).
- **$\textit{WealthQuintile}_i$** = Household wealth quintile rank of mother i (typically an ordinal scale from 1 = poorest to 5 = richest). Higher values indicate higher socioeconomic status.
- **$\textit{Education}_i$** = Education level of mother i (e.g. years of schooling completed). A higher value denotes more education.
- **$\textit{Urban}_i$** = Urban residence indicator for mother i (1 if living in an urban area, 0 if in a rural area). Controls for urban–rural differences in access and behavior.
- **$\textit{Age}_i$** = Age of mother i (in years). Included as a control.
- **$\textit{NumberOfChildren}_i$** = Number of children mother i has (parity). Controls for family size or prior births, which may affect healthcare utilization.

Regression Equation 4

$$\text{Numberofinfantdeaths}_i = \alpha_0 + \alpha_1 \text{WealthQuintile}_i + \alpha_2 \text{Education}_i + \alpha_3 \text{Urban}_i + \alpha_4 \text{Age}_i + \alpha_5 \text{NumberofChildren}_i + \varepsilon_i$$

- ***Numberofinfantdeaths_i*** = Total number of infant deaths mother *i* suffered.
- ***WealthQuintile_i*** = Household wealth quintile rank of mother *i* (typically an ordinal scale from 1 = poorest to 5 = richest). Higher values indicate higher socioeconomic status.
- ***Education_i*** = Education level of mother *i* (e.g. years of schooling completed). A higher value denotes more education.
- ***Urban_i*** = Urban residence indicator for mother *i* (1 if living in an urban area, 0 if in a rural area). Controls for urban–rural differences in access and behavior.
- ***Age_i*** = Age of mother *i* (in years). Included as a control.
- ***NumberofChildren_i*** = Number of children mother *i* has (parity). Controls for family size or prior births, which may affect healthcare utilization.

Regression Result

Table 1: Comparison of Survey-Weighted Regression Models

	ANC Visits (1) OLS (Linear)	Institutional Delivery (2) Logistic	Postnatal Check (3) Logistic	Infant Deaths (4) OLS (Linear)
Wealth: Poorer	0.056 (0.205)	0.416* (0.240)	-0.155 (0.231)	0.015** (0.008)
Wealth: Middle	0.554*** (0.212)	0.985*** (0.248)	0.134 (0.205)	0.011 (0.009)
Wealth: Richer	0.471* (0.274)	1.082*** (0.251)	0.228 (0.221)	0.016** (0.008)
Wealth: Richest	-0.014 (0.709)	1.751*** (0.514)	-0.191 (0.501)	0.011* (0.006)
Education: Primary	0.277* (0.161)	0.220 (0.239)	-0.096 (0.189)	0.023 (0.017)
Education: Secondary	0.453*** (0.164)	0.488** (0.203)	-0.198 (0.219)	0.057* (0.035)
Education: Higher	0.605** (0.246)	0.887*** (0.314)	0.171 (0.312)	0.017 (0.031)
Urban Residence	0.171 (0.117)	0.551*** (0.135)	-0.015 (0.097)	0.001 (0.005)
Mother's Age	0.037*** (0.011)	0.026** (0.012)	0.003 (0.010)	-0.003*** (0.0005)
Number of Children	-0.291*** (0.038)	-0.412*** (0.048)	-0.115*** (0.043)	0.009*** (0.002)
Province Dummies ^a	Included	Included	Included	Included
Interaction Terms ^b	Yes	Yes	Yes	Yes
Constant	3.194*** (0.312)	-0.834** (0.347)	-0.470* (0.265)	0.097*** (0.014)
Observations	4,006	4,857	4,001	26,028
Log Likelihood	-9,069.443	-	-	10,779.950
AIC	18,196.89	-	-	-21,501.90

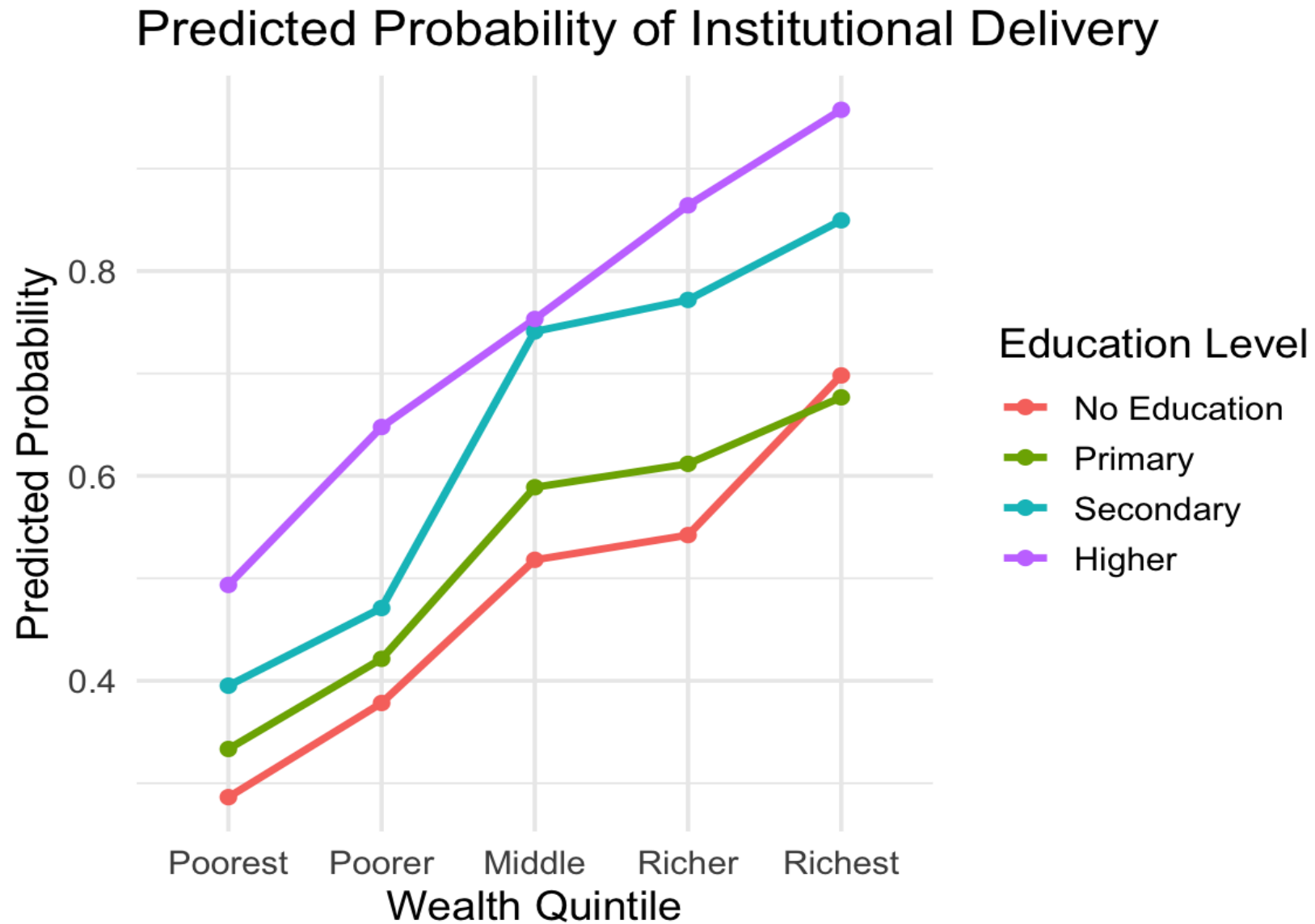
Notes: Robust standard errors in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

^a Province dummies v0242 to v0247 are included but not displayed for brevity.

^b All interaction terms between wealth and education categories are included.

Visualize Results?



Regression Result

Table 1: Fixed Effects and Wealth $\times$ Education Interactions Across Models

	ANC Visits (1) OLS (Linear)	Institutional Delivery (2) Logistic	Postnatal Check (3) Logistic	Infant Deaths (4) OLS (Linear)
Province 2	-0.271 (0.186)	-0.495*** (0.187)	-0.246 (0.150)	-0.005 (0.007)
Province 3	0.103 (0.198)	0.215 (0.204)	0.272* (0.164)	-0.003 (0.006)
Province 4	-0.229 (0.184)	0.250 (0.241)	0.106 (0.172)	-0.004 (0.006)
Province 5	-0.306** (0.146)	-0.134 (0.220)	-0.306** (0.154)	0.0005 (0.007)
Province 6	-0.792*** (0.174)	-0.242 (0.241)	-0.438* (0.238)	0.008 (0.010)
Province 7	-0.062 (0.149)	0.916*** (0.255)	0.592*** (0.171)	0.002 (0.009)
Poorer $\times$ Primary	-0.031 (0.270)	-0.040 (0.373)	0.236 (0.309)	-0.035* (0.018)
Middle $\times$ Primary	-0.192 (0.275)	0.067 (0.330)	-0.086 (0.307)	-0.029 (0.019)
Richer $\times$ Primary	0.209 (0.355)	0.065 (0.356)	-0.120 (0.343)	-0.040** (0.018)
Richest $\times$ Primary	1.291 (0.855)	-0.320 (0.591)	0.249 (0.610)	-0.018 (0.019)
Poorer $\times$ Secondary	0.245 (0.233)	-0.107 (0.285)	0.340 (0.320)	-0.058 (0.036)
Middle $\times$ Secondary	0.270 (0.250)	0.491 (0.314)	0.204 (0.291)	-0.071** (0.035)
Richer $\times$ Secondary	0.609* (0.310)	0.562* (0.303)	0.283 (0.288)	-0.072** (0.036)
Richest $\times$ Secondary	0.942 (0.732)	0.405 (0.514)	0.871 (0.529)	-0.061* (0.035)
Poorer $\times$ Higher	0.501 (0.349)	0.219 (0.393)	0.453 (0.408)	-0.045 (0.032)
Middle $\times$ Higher	0.645 (0.419)	0.156 (0.499)	-0.269 (0.404)	-0.043 (0.031)
Richer $\times$ Higher	0.827* (0.429)	0.793 (0.488)	-0.192 (0.442)	-0.020 (0.034)
Richest $\times$ Higher	2.655*** (0.915)	1.388** (0.701)	0.820 (0.585)	-0.030 (0.031)
Observations	4,006	4,857	4,001	26,028
Log Likelihood	-9,069.443	—	—	10,779.950
AIC	18,196.890	—	—	-21,501.900

Notes: Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
Only provincial fixed effects and wealth $\times$ education interaction terms shown.

Summarized Empirical Findings

Outcome	Wealth Effect	Education Effect	Interaction Effect	Geographic Effect
ANC Visits	Middle quintile: +0.55 (p<0.01) 4D 5D	Higher education: +0.61 (p<0.05) 4D 5D	Richest × Higher Ed: +2.66 (p<0.01) 4D	Province 6: -0.79 (p<0.01) 4D
Institutional Delivery	Richest: OR 1.75 (p<0.01) 4D 5D	Higher education: OR 0.89 (p<0.01) 4D 5D	Richest × Higher Ed: OR 1.39 4D	Province 2: -0.50 (p<0.01) 4D
Postnatal Care	Richest: -0.19 (NS) 4D	Higher education: +0.17 (NS) 4D	Richest × Higher Ed: +0.82 (NS) 4D	Province 7: +0.59 (p<0.01) 4D
Infant Deaths	Wealth: generally not significant 4D 5D	Higher education: -0.009 (p<0.05) 4D 5D	Middle × Secondary: -0.07 (p<0.05) 4D	Older age: -0.003 (p<0.01) 4D 5D

Key Findings:

- ❖ Wealth and education both independently increase the likelihood of institutional delivery and ANC visits, but their effects are multiplicative: women who are both wealthy and highly educated show the highest utilization rates.
- ❖ The effect of wealth on ANC visits is non-linear, with the strongest effect in the middle quintile; the richest quintile does not see a significant additional increase.
- ❖ Urban residence and older maternal age are associated with greater service utilization, while higher parity reduces it.
- ❖ Provincial disparities are substantial, with some provinces lagging significantly behind others even after controlling for SES.
- ❖ For infant mortality, direct wealth effects are limited, but higher education and its interaction with wealth are associated with lower infant deaths, suggesting education's protective effect transcends wealth.

What does Policy say about our Findings?

Targeted Economic Interventions

- ❖ Aama Program (2009–present): Provides cash transfers for ≥ 4 ANC visits (NPR 800), institutional delivery (NPR 1,000–3,000), directly addressing wealth barriers identified in Table 1 (wealth quintile effects: Richest vs. Poorest OR = 5.77 for institutional delivery).
- ❖ Transportation subsidies (NPR 500–1,500) for remote areas, mitigating rural disadvantages (rural residence reduced ANC visits by 0.23 in Table 1).
- ❖ Social Health Insurance Act (2017): Covers maternity services under universal health coverage, reducing out-of-pocket costs that disproportionately affect poorer households (aligned with insignificant wealth effects in infant mortality models).

Educational and Awareness Campaigns

- ❖ School Health and Nursing Service Program: Integrates adolescent sexual/reproductive health education, addressing the education gradient where higher education increased ANC visits by 1.7 (Table 1). Explains why secondary/higher education showed stronger effects than primary education in regression models.
- ❖ ANC Counseling Protocols: Mandates counseling during the first ANC visit to promote follow-ups, countering parity-related declines (each child reduced ANC visits by 0.31 in Table 1).

Persistent Gaps and Future Directions

While policies align with observed wealth/education gradients and provincial disparities, challenges remain:

- ❖ Indirect costs (e.g., lost wages) limit poorest quintile participation despite cash transfers.
- ❖ Postnatal care policies lack the rigor of ANC/delivery incentives, reflected in weaker wealth-education interactions for postnatal checks (Table 2).
- ❖ Madhesh Province requires targeted cultural competency training to address residual access barriers.

Implications:

- ❖ Policy interventions should prioritize both economic (e.g., cash transfers, transportation subsidies) and educational (e.g., female education, health literacy campaigns) dimensions. (wealth multiplies in effectiveness when paired with human capital like education)
- ❖ Addressing indirect costs (lost wages, childcare, travel) is essential, as current cash transfers do not fully offset these barriers for the poorest.
- ❖ Provincial targeting and context-specific strategies are required to close persistent geographic gaps.
- ❖ Integrated, multidimensional approaches-rather than isolated economic or educational interventions-are likely to yield the greatest improvements in maternal and child health outcomes